

NATIONAL ECONOMICS UNIVERSITY

Faculty of Data Science and Artificial Intelligence

MACHINE LEARNING

Machine Learning-Based Electricity Consumption Forecasting in Tetouan City

A chronological evaluation of conventional regression models

GROUP MEMBERS

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Can machine learning beat a strong daily pattern?

PROBLEM MOTIVATION & RESEARCH QUESTIONS

Electricity planning needs the **shape of tomorrow's hourly demand**.

The practical reference

If demand repeats every day, copying yesterday is already competitive.

1. How much do conventional models improve on seasonal baselines?
2. Which feature groups add predictive value?
3. How do tuning and the target formulation affect the result?

Evaluate future periods and interpret gains against the named baseline.

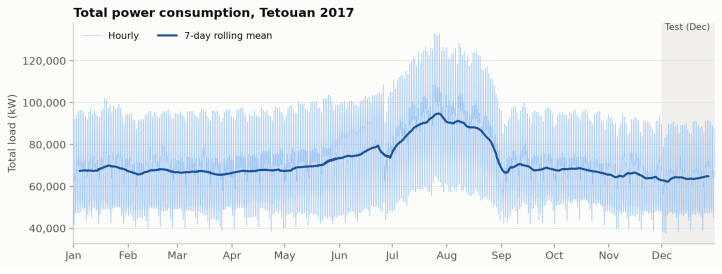
The target is mean hourly power across three zones

$$y_t = \frac{1}{6}\sum_{j=0}^{5}\left(Z_{1,t,j} + Z_{2,t,j} + Z_{3,t,j}\right) \quad [\mathrm{kW}]$$

52,416 ten-minute records

8,736 hourly means

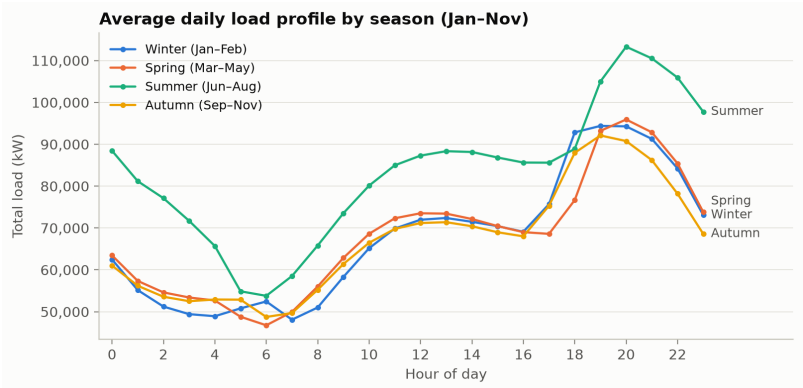
1 Jan–30 Dec 2017, Tetouan



Mean power (kW); hourly energy is $y_t \times 1\,\mathrm{h}$ (kWh).

Whole-year view includes December, but no 31 December data; the centred smooth is descriptive.

Daily shape and season both matter



JANUARY–NOVEMBER

06–07h
daily minimum

19–20h
evening peak

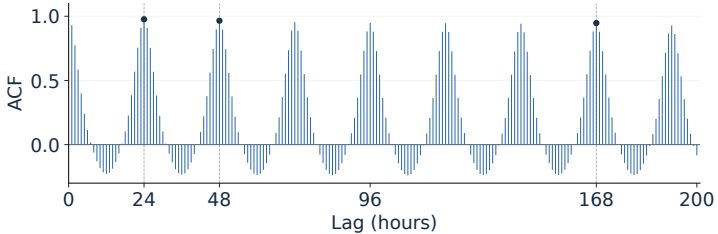
Summer
higher load profile

Use calendar information and past load together.

Profiles use January–November development data.

Today looks like yesterday

JANUARY–NOVEMBER DATA ONLY



ACF: **24 h = 0.98**, **48 h = 0.96**, **168 h = 0.95** (weekly peak).

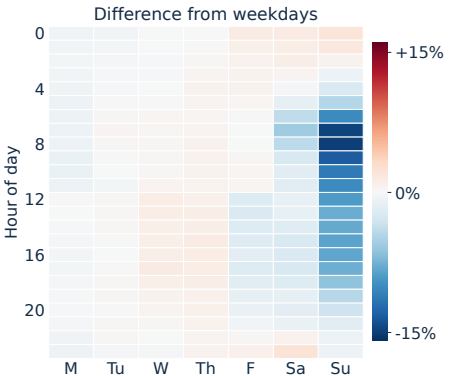
Lag 12 h is negative: day and night are in opposite phases.

Copying yesterday (Naive-24h) is therefore a strong baseline.

→ **features:** `lag_24`, `lag_48` **(daily history).**

Sundays and Mondays break the daily pattern

JANUARY–NOVEMBER / DIFFERENCE FROM MON–FRI AT THE SAME HOUR



Sundays: **5.9% lower** on average; about **15% lower at 7–8 h**.

Saturday mornings: **4–6% lower**.

The effect varies by hour; a nonlinear model can learn this interaction.

Copying yesterday mismatches **Monday/Sunday** and **Sunday/Saturday**.

→ **features:** `dayofweek`, `is_weekend`, **cyclic weekday.**

h24 and h1 use different information at issuance

h24: midnight batch

00:00 on day D: forecast its 24 hourly intervals.

History: $y_{t-24},\ y_{t-48}$.

Lead to interval start: 0–23 h.

Lead to interval completion: 1–24 h.

h1: hourly update

Update each hour for the next complete hourly interval.

History: $y_{t-1}, y_{t-2}, y_{t-3}, y_{t-24}$.

Uses the latest completed-hour load.

Prompt measurements assumed.

The 24-hour lag is a minimum history offset; lead times vary across the daily batch.

Both settings assume prompt load measurements and use observed target-hour weather.

Seventeen inputs combine timing, weather and history

MAIN h24 MODEL: FEATURE SET D

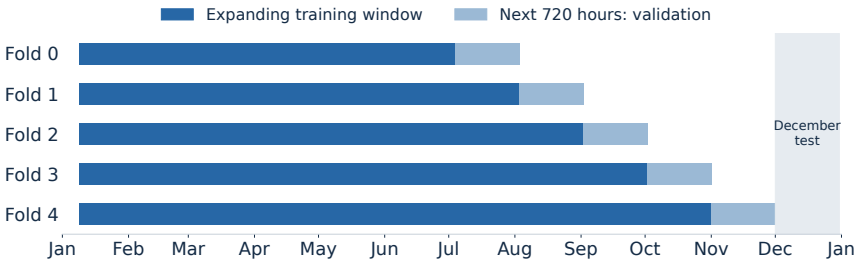
Group	Inputs	Count
Weather	Temperature, humidity, wind speed, general diffuse, diffuse	5
Calendar	Hour, day of week, weekend; sine and cosine for hour and weekday	7
Special days	Religious holiday, civil holiday, Ramadan indicators	3
Daily history	y_{t-24}, y_{t-48}	2

Target-hour observed weather requires forecasts at issuance.

History shifted; scalers fitted per fold. Weekly/rolling inputs excluded from set D.

Model selection stays within January–November

JAN–NOV DEVELOPMENT: 7,825 HOURS **DECEMBER TEST: 720 HOURS**



After history warm-up, five expanding windows each validate the next 30 days.

Selection: lowest mean CV RMSE → refit on all development data → December.

Tuning and reporting reuse folds (no nested CV); zero gap assumes prompt data.

Learn the load level—or a correction to yesterday

TWO SEASONAL BASELINES

$$\widehat{y}_t = y_{t-24} \qquad \widehat{y}_t = y_{t-168}$$

FIVE REGRESSION FAMILIES

Linear regression / tuned Ridge
Decision tree (DT)
Random forest (RF)
RBF support vector regression (SVR)
XGBoost (XGB)

TWO TARGET FORMULATIONS

Absolute: learn y_t.

Residual (Δ):

$$r_t = y_t - y_{t-24}$$

$$\widehat{y}_t = y_{t-24} + g(\mathbf{x}_t)$$

Learn the change from yesterday so recent load carries the current level.

Both target formulations compete on development CV.

RF averages are range-bounded; boosted trees do not share this strict bound.

Why predict a correction? July tests extrapolation

FOLD 0: 4 JULY–2 AUGUST / TUNED MODELS

A new load range

Training maximum: **110,711 kW**
Validation maximum: **133,130 kW**
About **12%** of hours exceed the training maximum.
For $r_t = y_t - y_{t-24}$, **0%** fall outside the training residual range.

July-fold RMSE (kW)

Model	Absolute	Residual
RF	5,554	**3,215**
XGBoost	5,344	**2,979**

RF averages cannot extrapolate beyond training targets. Learning a correction reduces the impact of the level shift.

Residual learning is not necessary for the selected SVR.

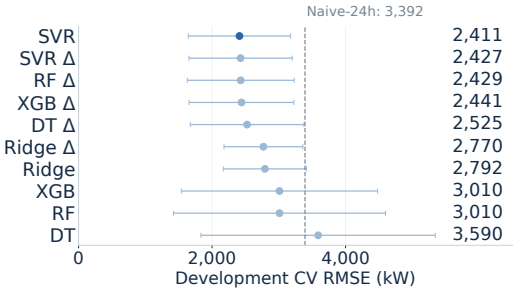
SVR CV: absolute **2,411** vs residual **2,427 kW**; nearly tied.

Target variants were tuned separately; gains also reflect their chosen parameters.

Absolute SVR leads CV, with close alternatives

Randomized search: **170 configurations** × **5 temporal folds per target type**;

SVR CV RMSE fell **39.3%** versus default SVR. Bergstra & Bengio (2012).



2,410.6

kW mean CV RMSE

± 766.3 kW fold SD.

Default models: XGB wins November; LR (**2,792 kW**) wins five-fold CV.

SVR and DT trail Naive-24h (**3,392 kW**) on CV.

More recent history: h24 vs h1 (CV). SVR 2,410.6 → 1,866.4 kW (-22.6%).

Same configurations/dates; histories and initial rows differ. CV folds are reused.

Daily lags drive accuracy; the expanded set hurts

ABSOLUTE-TARGET RF / FIVE-FOLD CV / FIXED D-TUNED PARAMETERS

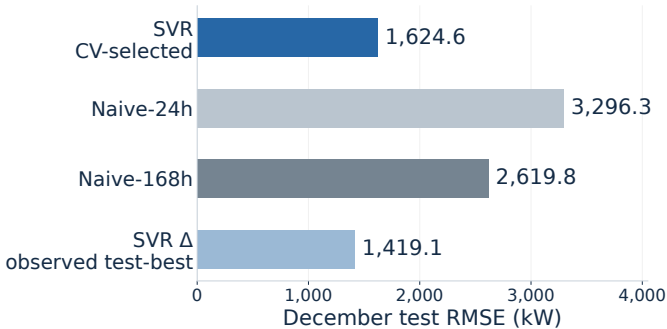
Set	Feature group	Inputs	RMSE (kW)	Change
A	Weather only	5	15,010	—
B	+ Calendar	12	10,041	−33.1%
C	+ Special days	15	9,168	−8.7%
D	**+ Daily lags (24 h, 48 h)**	**17**	**3,010**	−67.2%
E	+ Weekly lag and rolling summaries	21	3,700	+22.9%

D is selected. Daily lags give the largest gain; adding weekly/rolling features worsens RF.

XGBoost also worsens: **3,010 → 4,921 kW** (D → E).

Possible mechanism: rolling levels may hinder July extrapolation. D/E does not isolate the weekly lag. E/F weather removal changes RMSE by up to 5.3%; this is not an SVR test.

The selected SVR halves daily-baseline December RMSE



50.7%

RMSE reduction vs Naive-24h

38.0% vs Naive-168h

MAE: **1,094.6 kW**

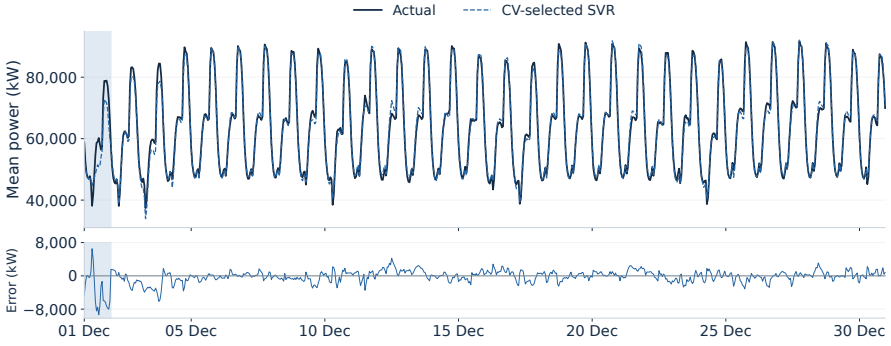
MAPE: **1.78%**

R^2: **0.9869**

Residual SVR: observed test-best; not CV-selected. Absolute SVR remains final.

December 1–30 only; results condition on observed target-hour weather.

Good daily tracking still leaves concentrated errors



Bias: -202.1 kW

Error = forecast − actual.

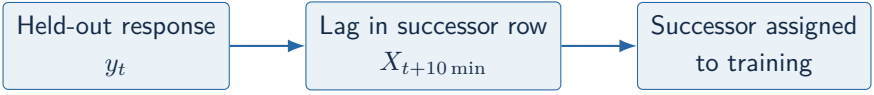
Worst day: 1 December (Mawlid)

MAE 4,579.2 kW; highlighted in blue.

Possible explanation: the holiday flag may transfer larger Eid demand reductions to Mawlid.

Evaluation shortcuts change what the score means

A CONCRETE LEAKAGE MECHANISM IN THE SHUFFLED DIAGNOSTIC



Separate RF tuning diagnostic (20 candidates; errors in kW)

CV strategy	CV estimate	December test
Shuffled KFold	2,007.6	2,173.6
Expanding TimeSeriesSplit	3,012.4	2,044.2

Same Random Forest: shuffled 10-min split R^2 0.999 vs chronological h24 R^2 0.980.

Forecast settings, features and test sets differ; this is a descriptive comparison.

Findings & limitations

FINDINGS

1. Choose the final model on chronological CV; report December separately.
2. Daily lags give the largest RF gain; weekly/rolling features hurt.
3. The CV-selected SVR halves December Naive-24h error (-50.7%).

LIMITATIONS

- Weather is observed, not forecast; prompt recent-load measurements assumed.
- One city, one year, one 30-day test.

Next: issuance-time weather and independent seasonal tests.

Conclusion & Q&A

CV-SELECTED ABSOLUTE-TARGET SVR / DECEMBER 2017

1,624.6 kW
RMSE on 720 hourly observations

50.7%
lower RMSE than Naive-24h

A promising chronological benchmark, conditional on observed weather.

Thank you. Questions?